Local primitivity in finite flag-transitive linear spaces

Human Compression project: private unreviewed edition

Abstract

We give a corrected classification of the faithful flag-transitive linear-space actions considered by Buekenhout, Dehaye and Leemans. The large classifications used by that paper remain named external inputs. The accompanying common paper proves the local-action, field, orbit and semilinear descent arguments; this application supplies the complete case assembly, the exact acting-group converses and the finite Feit bound. Several literal source assertions require corrections. Thus the result is a clausewise adjudication of the complete source contract, not a claim that every printed theorem follows unchanged.

1 Objects, source and retained classification inputs

A linear space here is finite, has a unique line through two distinct points, at least two points per line, and at least two lines per point. The group $G \leq \mathrm{Aut}(\Gamma)$ is faithful and flag-transitive; it need not be the full automorphism group. Write v, k, r for point number, line size and pencil size. Flag transitivity makes k, r constant and $r = (v-1)/(k-1)$. We call the action locally primitive when G_x is primitive on its r lines and G_L is primitive on its k points, taking the actual induced actions and their kernels. For the source's definition of residual weak primitivity (RWPRI), this is equivalent by the retained Higman–McLaughlin point-primitivity theorem; the primitive-stabilizer criterion is the standard Dixon–Mortimer fact. Local two-transitivity means two-transitivity of both these actions and is the source's $(2T)_1$ condition.

The primary edition is the complete twenty-two-page article [1], printed pages 25–46. Its bibliography and theorems quoted from earlier authors are part of its dependency boundary. We retain the following inputs, at their actual permutation-representation scope. Their large classification proofs are not claimed as new proofs here.

1. BDDKLS, with the subsequent classification work explicitly listed by the source (Liebeck, Delandtsheer, Kleidman, Saxl and the sporadic cases), reduces a thick non-one-dimensional flag-transitive linear space to a projective space, affine space, one of the two Hering spaces, a Witt–Bose–Shrikhande space, a Hermitian unital or a Ree unital [1, Proposition 1.1 and its attribution].

2. Kantor and Cameron–Kantor's projective classification gives the classical special-linear case, the specified A_7 action on $\mathrm{PG}(3, 2)$, or the sharply flag-transitive Frobenius plane. Feit's exact arithmetic restrictions on the last case are retained [1, Propositions 3.1–3.2].

3. For affine dimension at least three, retain the precise BDDKLS–Huybrechts alternatives and their natural field-module identifications: the one-dimensional case; the impossible last-derived-$2.A_5$ case on $\mathrm{AG}(4, 3)$; or a point-two-transitive action with one of the six representation types in source Proposition 3.4. The normal natural groups there are $\mathrm{SL}_u(q^{n/u})$, $\mathrm{Sp}_u(q^{n/u})$ or $G_2(q^{n/6})'$, together with the three listed small exceptions. Their invariant extension-field structures are part of these representation identifications, not an inference from an abstract isomorphism type.

4. Retain Foulser's actual affine-plane representations. These include the ten groups on $\mathrm{AG}(2, 9)$ with common minimal subgroup $3^4 : 2.A_5$. The induced special-linear group is the projective special-linear group, not necessarily the full projective linear group. The two printed inclusions are corrected below.

5. The BDD non-one-dimensional non-Desarguesian planes are the Lüneburg planes, the nearfield plane of order nine and the Hering plane of order twenty-seven. Retain their specified full-group failures: Lüneburg is not locally point-primitive; the latter two are not locally line-primitive. Retain the two 3^6-point Hering spaces of line size nine with group $3^6 : \mathrm{SL}_2(13)$, locally primitive but not locally two-transitive; and the Ree-unital full-group failure of local line-primitivity [1, §§3.2.2, 3.3, 3.6].

6. Retain the full automorphism identifications for the WBS and Hermitian geometries, respectively $\mathrm{P\Gamma L}_2(2^e)$ and $\mathrm{P\Gamma U}_3(q)$, with the classical natural models. Their required local actions are proved in the common paper, including the small-parameter exceptions to the source assertions.

7. For circles, retain the complete almost-simple and affine point-two-transitive representation roster quoted as source Proposition 4.1, with the original exact Foulser exception list. Also retain the specified Mathieu transitivity, Atlas stabilizer/action identifications for the sporadic and small alternating cases, and the classical orthogonal-point symplectic subdegrees. These local representation inputs are stated explicitly in Section 3.

8. Delandtsheer's one-dimensional reduction supplies all translations and the spread/nonspread dichotomy. Retain the full unordered intersecting-line-pair classification in source Proposition 5.1(ii), the nonspread divisibility in Proposition 5.7 and the more general prime-power repunit theorem quoted from Ribenboim in Proposition 5.3. Our own prime-pencil proof uses only the elementary prime case and is written in full; it does not reprove the quoted prime-power theorem. For precision, the retained unordered-pair exceptions are: the circles with $\mathrm{AGL}_1(3)$, $\mathrm{AGL}_1(4)$, $A\Gamma L_1(4)$ or $A\Gamma L_1(8)$; the Fano plane with $7:3$; and $\mathrm{AG}(2,4)$ with $A\Gamma L_1(16)$. The repunit statement retained is that $(x^u-1)/(x-1) = r^m$, $u \geq 3$, $x > 1$, r prime, forces u prime and the multiplicative order of x modulo r equal to u; if $x = s^b$ as in the quoted statement, then $b = u^e$ and $r \equiv 1 \pmod{u^{e+1}}$, with r not a Fermat prime. Its prime specialization below is separately proved.

The natural G_2 module and maximal parabolic identification are retained classical building inputs. Its binary derived-group boundary uses one complete small GAP maximal-subgroup calculation. The large classification inputs, missing original software and nontext graphics have no asserted zero cost or full formalization.

2 Thick actions and the affine comparison

The common local-action criterion identifies a translation pencil with its spread, and the line action with $U \rtimes H_U|_U$. Thus the two conditions are precisely pencil primitivity and irreducibility of the *actual restriction* $H_U|_U$; for local two-transitivity they are pencil two-transitivity and transitivity on $U \setminus \{0\}$ [2, Proposition 1.2]. The distinction matters when a line stabilizer has a restriction kernel.

Classical projective and affine special-linear actions are locally two-transitive by the explicit basis and determinant arguments in [2, Proposition 2.1]. In the exceptional projective A_7 action the retained local groups are $\mathrm{PSL}_3(2)$ on seven pencil points and S_3 on a line; both are two-transitive. The other projective branch reduces, by the written Feit congruence argument, to the Fano plane with $7:3$; its cyclic local groups of order three are primitive but not two-transitive [2, Proposition 5.1].

For the affine classification one cannot assume that two geometries are nested merely because their line sizes are ordered. The full replacement for source Lemma 3.5 is [2, Theorem 6.5]: a primitive affine pencil has largest line size among all invariant affine spaces for the same permutation group. The proof treats non-Desarguesian planes as well: the SL and Sp vector stabilizers have outside orbits larger than the affine bound $\sqrt{v}$; the G_2 case uses its calculated commuting field and the named BDD Lüneburg exclusion. For an alternative line through two fixed points, uniqueness

makes it invariant under their stabilizer. The argument neither postulates a common field in advance nor identifies G with a full automorphism group.

More directly, extension-field normalizers preserve the actual coarser field-line spread. The commuting-algebra lemma therefore excludes all proper-field natural modules from a primitive pencil. The remaining natural symplectic pencil is primitive and has two nontrivial perpendicularity orbits. Its whole normalizer preserves that relation, so no outer extension makes it two-transitive [2, Proposition 2.2]. The same normalizer argument and the retained maximal-parabolic action treat $G_2(q)$, $q \geq 4$ even [2, Proposition 6.3]. For $q = 2$ use [2, Proposition 4.4], with the required complete order list $6048; 96, 96, 168, 216$ and automorphism order 12096. Its degree-63 pencil is primitive and not two-transitive.

The other Huybrechts cases have imprimitive pencils by the explicit normal-subgroup/order exclusions in the common paper. The Foulser plane exceptions reduce to $q = 9$: a faithful primitive A_4 or S_4 action has degree at most four, and the primitive degrees of A_5 are $5, 6, 10$. The retained ten degree-81 groups are all locally primitive. In their minimal $2.A_5$ linear group an order-four lift f of an involution has $f^2 = -I$ and fixes a vector line over $\mathbb{F}_9$; its order-four scalar acts irreducibly on that two-dimensional $\mathbb{F}_3$-line. This supplies the line-primitivity proof without the source's unjustified assertion that the whole Sylow subgroup fixes the same directions [2, Proposition 2.4]. Their pencil images A_5, S_5 cannot be two-transitive of degree ten by order divisibility.

The retained non-Desarguesian-plane and Ree exclusions pass from full groups to every subgroup: a primitive subgroup would force primitivity of the full local action. The Hering spaces retain precisely their stated BDD local properties. For WBS spaces the uniform common proof is stronger than a finite check but corrects the source's stated reason: if $q + 1$ is composite the pencil is imprimitive; if it is prime, e is a power of two and the line action on $\mathbb{F}_{2^e}/\mathbb{F}_2$ has reducible cyclic two-group stabilizer. Thus no WBS space is locally primitive. At $q = 8$ the line action of the full group is A_4 of degree four and *is* primitive; its nine-point pencil fails instead [2, Propositions 8.1–8.2].

For the Hermitian unital, put $d = \gcd(3, q + 1)$ and $\mathcal{C} = \mathbb{F}_{q^2}^*/(\mathbb{F}_{q^2}^*)^d$. The common matrix calculation defines the exact outer action $\pi : G/\operatorname{PSU}_3(q) \to \operatorname{Sym}(\mathcal{C})$. The pencil group in the special-unitary case is $\mathbb{F}_{q^2}^+ \rtimes (\mathbb{F}_{q^2}^*)^d$. For $q > 2$ it is primitive; for $q = 2$ it is not. In either case the full acting group is locally two-transitive exactly when $\pi(G)$ is transitive. For $q = 2$ this same condition is necessary for local primitivity. All line actions are already two-transitive in the special-unitary subgroup [2, Theorem 7.1]. In particular the literal blanket converse for every $\operatorname{PSU}_3(q)$-containing group is false.

3 The complete corrected circle test

For a circle, flags are ordered pairs of distinct points. Thus flag transitivity is point two-transitivity, RWPRI is primitivity of the point stabilizer on the remaining points, and $(2T)_1$ is point three-transitivity [2, Proposition 1.4]. The following list is necessary and sufficient within the retained two-transitive representation classification. Repeated or overlapping small actions are permitted; the entries describe actions, not just group names.

Locally primitive circle action	Locally two-transitive exactly when
A_v or S_v, $v \geq 5$	Always.
$\operatorname{PSL}_2(q) \leq G \leq \operatorname{P\Gamma L}_2(q)$, $v = q + 1$	q even, or an allowed semilinear representative has nonsquare determinant.
$\operatorname{Sp}_{2n}(2)$, $v = 2^{2n-1} \pm 2^{n-1}$, $n \geq 3$	Never.
$\operatorname{PSL}_2(11)$, $v = 11$	Never.
M_v, $v = 11, 12, 22, 23, 24$; $\operatorname{Aut}(M_{22})$, $v = 22$; M_{11} in degree twelve	Always.

Locally primitive circle action	Locally two-transitive exactly when
HS in degree 176; Co_3 in degree 276	Never.
$2^n : \mathrm{GL}_n(2)$, $n \geq 2$	Always.
$2^n : H$, $n \geq 4$ even, with natural $\mathrm{Sp}_n(2) \leq H \leq N_{\mathrm{GL}_n(2)}(\mathrm{Sp}_n(2))$	Never.
$2^4 : A_6$ and $2^4 : A_7$ in their retained natural binary actions	Exactly the A_7 case.
$2^6 : H$, $G_2(2)' \leq H \leq N_{\mathrm{GL}_6(2)}(G_2(2)')$	Never.
One-dimensional, point-two-transitive $G \leq A\Gamma L_1(v)$, with $v-1$ prime	Exactly $v = 3, G = S_3$, or $v = 4, G = S_4$.

Here are the complete local justifications, including the retained small-action boundaries. Alternating groups in the stated range are three-transitive by even-permutation completion of a prescribed triple; for A_5 the two unused letters already suffice. The projective-line entry follows from the scalar-square quotient calculation in [2, Proposition 4.2]; it includes mixed field-diagonal extensions and does not require containment of $\mathrm{PGL}_2(q)$ when that would be too strong. The source's formulation via normalization of a sharply three-transitive group is retained as the attributed Gardiner–Praeger–Zhou description of these same extensions; our determinant criterion tests the actual specified action.

For the symplectic orthogonal-point actions, retain the rank-three subdegrees and use the complete block-size divisibility proof in [2, Proposition 4.5]. For $\mathrm{PSL}_2(11)$ the retained point stabilizer is A_5 on ten points, with stabilizer S_3. The elementary maximality proof in [2, Proposition 2.4] gives primitivity, while $10 \cdot 9 \nmid 60$ rules out two-transitivity. The stated Mathieu transitivity is the retained exact permutation-action input. In the HS action the retained Atlas point stabilizer is $\mathrm{PSU}_3(5) : 2$, primitive but not two-transitive on the other175 points. In the Co_3 action the retained two-point inclusion $\mathrm{PSU}_4(3) : 2 < McL : 2$ is maximal and its degree-275 action is not two-transitive. These are explicit retained local-action claims, not order-only inferences.

For binary affine groups, translations reduce the question to the nonzero-vector action of the point stabilizer. The natural linear and symplectic cases have already been proved in the common paper. The retained $A_6 \leq \mathrm{GL}_4(2)$ action has point stabilizer S_4 maximal of index fifteen; it is primitive, and $15 \cdot 14 \nmid 360$ excludes two-transitivity. The retained $A_7 \leq \mathrm{GL}_4(2)$ action is the two-transitive action on the fifteen points of $\mathrm{PG}(3,2)$, giving three-transitivity of its affine extension. This does not make the *fifteen-point circle* of A_7 primitive: there the point stabilizer preserves seven pairs, the punctured projective lines. The binary $G_2(2)'$ case is the exact small case proved above.

In the one-dimensional entry, the point stabilizer is cyclic by cyclic. Its primitive remaining-point action therefore has prime degree $r = v - 1$, and this condition is sufficient by prime-degree transitivity. Two-transitivity on those r points requires $r(r-1) \mid rd$ for $v = p^d$, so $p^d - 2 \mid d$. For $d \geq 3$, $2^d - 2 > d$; for $d = 2$ only $p = 2$ works, and for $d = 1$, $v \geq 3$ leaves only $p = 3$. The required group orders force S_4 and S_3, respectively, and both suffice.

For exhaustion, run through the retained roster in source Proposition 4.1. Higher-dimensional projective and unitary actions preserve actual thick linear spaces; their punctured lines form nontrivial invariant pencil partitions. Suzuki and Ree point stabilizers have regular unipotent groups with proper nontrivial characteristic centers, giving local block systems even in the outer extensions. Odd-characteristic affine modules of dimension at least two preserve thick prime-field affine lines. For even extension fields larger than two the actual normalized extension-field lines give the same exclusion. The remaining extraspecial and $\mathrm{SL}_2(5)$ or $\mathrm{SL}_2(13)$ affine exceptions have odd characteristic and are excluded this way. Over the binary field the cases in the table remain. This accounts for every branch of the retained roster. The small circles are included exactly: degree three requires S_3; degree four permits A_4, S_4 for local primitivity and only S_4 for local two-transitivity. Literal extensions of the source's alternating-group footnotes to A_3 or to a three-transitive A_4 are invalid.

4 One-dimensional spaces and the acting-group converse

In the spread case, the complete semilinear descent proof in [2, Theorem 3.3] gives a Desarguesian spread with

$$v = p^{u^{e+1}}, \qquad k = p^{u^e}, \qquad r = (v-1)/(k-1) \text{ prime}, \qquad p, u \text{ prime}, \quad e \geq 0.$$

It proves the earlier source question affirmatively at this precise one-dimensional, flag-transitive, locally primitive scope. That answer is reused from the accepted phase-four proof, with no new question credit. For a specified acting group $G = F^+ \rtimes H \leq A\Gamma L_1(F)$, flag transitivity and the prime pencil do *not* alone suffice. The exact further condition is that $H_K|_K$ be irreducible over $\mathbb{F}_p$, for the line $K = \mathbb{F}_{p^{u^e}}$ through zero. This is both necessary and sufficient by the translation criterion. Existence for every allowed parameter set is proved using the actual one-dimensional Singer affine group $F^+ \rtimes F^*$, rather than borrowing the full higher-dimensional automorphism group.

An explicit counterexample to the source's unrestricted acting-group converse is $F = \mathbb{F}_{16}$, $K = \mathbb{F}_4$ and $G = F^+ \rtimes C_5$, with $C_5 \leq F^*$. The five K-directions form a regular C_5 orbit, so G is flag-transitive on $\mathrm{AG}(2,4)$ and satisfies the parameters of source Theorem 6.1(1). But $C_5 \cap K^* = 1$, so its action on a line is just the regular additive group $K^+ \cong C_2^2$. It preserves the two cosets of any order-two subgroup and is imprimitive. This corrects the printed converse, without affecting the existential assertion of Proposition 5.5.

The local-two-transitive spread cases are exactly $\mathrm{AG}(2,2)$ with $A\Gamma L_1(4)$ and $\mathrm{AG}(2,4)$ with $A\Gamma L_1(16)$; all necessity, uniqueness of the acting group and sufficiency proofs are in [2, Proposition 3.4]. The nonspread proof [2, Proposition 5.2] leaves exactly the Fano plane with $7:3$ and the three-point circle with S_3; only the latter is locally two-transitive. The source's Proposition 5.8 and its footnote removing the spread restriction miss that small circle when passing from $r = k$ to a projective plane. Both final source unions already contain it through the degree-three projective-line circle, so no additional family beyond their overall universe is needed.

5 The final classification with both directions

Theorem 5.1 (Corrected locally primitive classification)**.** *Subject to the named retained classifications above, the faithful flag-transitive locally primitive finite linear-space actions are exactly the following union, with their specified natural representations.*

1. *The one-dimensional Desarguesian spread actions of Section 4, with prime pencil and the actual line-stabilizer irreducibility condition.*
2. $\mathrm{PG}(n,q)$, $n \geq 2$, $\mathrm{PSL}_{n+1}(q) \leq G \leq \mathrm{P\Gamma L}_{n+1}(q)$*; the specified* A_7 *action on* $\mathrm{PG}(3,2)$*; and the Fano plane with* $7:3$*.*
3. $\mathrm{AG}(n,q)$, $q \geq 3$*, with translations and normal natural* $\mathrm{SL}_n(q)$, $n \geq 2$*, or* $\mathrm{Sp}_n(q)$, $n \geq 4$ *even, in its linear stabilizer.*
4. $\mathrm{AG}(6,q)$*,* q *even, with translations and normal natural* $G_2(q)'$ *in its linear stabilizer, including* $q = 2$*.*
5. *The ten specified Foulser groups on* $\mathrm{AG}(2,9)$ *with pencil image* A_5 *or* S_5*.*
6. *The two retained Hering spaces on* 3^6 *points with line size nine and group* $3^6 : \mathrm{SL}_2(13)$*.*
7. *A Hermitian unital, with* $\mathrm{PSU}_3(q) \leq G \leq \mathrm{P\Gamma U}_3(q)$*: all such actions for* $q > 2$*; for* $q = 2$ *precisely those whose scalar-coset action* $\pi(G)$ *is transitive.*
8. *The circle actions in the complete table of Section 3.*

All stabilizers in the affine entries are actual semilinear geometry stabilizers normalizing the stated natural subgroup. The entries can overlap; no claim of pairwise disjointness is made.

Proof. For a thick non-one-dimensional space apply the retained six-family BDDKLS reduction. The projective local calculation gives entry 2. The full affine comparison and the explicit pencil exclusions leave entries 3–5; the non-Desarguesian-plane cases fail one of the retained full-group local tests. The retained Hering local actions give entry 6. The uniform WBS proof and retained Ree exclusion remove those two families. The unitary scalar-coset calculation gives entry 7, with its necessary small-parameter qualification. The one-dimensional spread and nonspread arguments give entry 1 and the already listed Fano/three-point cases. Circles give exactly entry 8 by the exhaustion proof in Section 3. These arguments prove necessity.

Conversely, the translation criterion plus the explicitly stated restriction irreducibility proves entry 1. The basis, Frobenius, symplectic, exceptional-pencil and order-four-lift proofs in the common paper give entries 2–5 for every stated group; supergroups preserve local primitivity. Entry6 uses precisely its named retained BDD local-action assertion. The unitary calculation gives entry 7 for exactly the listed groups. Section 3 proves every circle entry. Each action is faithful and flag-transitive in its specified model; global point-primitivity is supplied by the retained Higman–McLaughlin theorem or by the direct translation proof. Thus both directions and every local hypothesis are established. □

Theorem 5.2 (Corrected locally two-transitive classification)**.** *Under the same retained inputs, the locally two-transitive actions are exactly the following subunion of Theorem 5.1:* $\mathrm{AG}(2,4)$ *with* $A\Gamma L_1(16)$*; the classical projective actions and the projective* A_7 *action; the natural affine special-linear actions with* $q \geq 3$*; the Hermitian actions for which* $\pi(G)$ *is transitive; and exactly the circle entries marked locally two-transitive in the table.*

Proof. Local two-transitivity implies local primitivity, so start with the complete preceding union. The one-dimensional proof leaves only the four- and sixteen-point cases, and the former is already a circle. The classical projective and affine actions and projective A_7 action are locally two-transitive by their explicit local groups. The Fano cyclic action, symplectic perpendicularity relation, exceptional G_2 relation/order bound, Foulser degree-ten order test and retained Hering local assertion exclude all other thick entries except unitals. There the necessary and sufficient condition is exactly the calculated transitivity of $\pi(G)$. The circle criterion and table finish both directions. This proves the classification for the specified acting groups, including the converses that were only asserted in the source. □

6 The finite Feit bound stated in the source

The source's physical page 5 reports a computer-assisted extension to plane order $m \leq 10^{10}$ of Feit's exclusion. This is a separate finite assertion, not a consequence of local primitivity, and we retain it explicitly. The following calculation recovers the stated bound while using Feit's necessary conditions as an attributed external theorem.

Proposition 6.1 (The declared finite range)**.** *Assume the retained Kantor–Cameron–Kantor classification and Feit's theorem in source Propositions 3.1–3.2. A flag-transitive projective plane of order* $m \leq 10^{10}$ *is Desarguesian.*

Proof. The retained classification reduces a possible non-Desarguesian example to its sharply flag-transitive Frobenius case. Feit's necessary conditions then give

$$8 \mid m, \quad m \text{ not a power of 2}, \quad v = m^2 + m + 1 \text{ prime}, \quad d^{m+1} \equiv 1 \pmod{v} \quad (d \mid m).$$

The order of m modulo v is exactly three: $m^3 \equiv 1$ and $1 < m < v$ rules out order one. The condition with $d = m$ therefore gives $3 \mid m + 1$. Combining with $8 \mid m$ gives $m = 8 + 24t$. The condition with $d = 2$ gives $2^{m+1} \equiv 1 \pmod{v}$.

The complete required program `feit_bound.cpp` checks all 416 666 667 integers of the form $8 + 24t$ in the declared range. It generates the primes $p \leq 10^4$ by the ordinary Eratosthenes sieve and enumerates, directly, all residue classes $t \bmod p$ for which $(8 + 24t)^2 + (8 + 24t) + 1 \equiv 0$

(mod p). Segmented marking rejects a value only when that p is a proper divisor of v; the small possible equality $v = p$ is explicitly retained. Thus every marked rejection contradicts the required primality, with no probabilistic primality test. Powers of two are separately excluded by the exact bit test. On every remaining value, GMP computes v and $2^{m+1} \bmod v$ as arbitrary-precision integers. The complete saved result is

proper-divisor rejections	331 513 606
excluded powers of two	16
remaining exact modular powers	85 153 045
total	416 666 667.

None of the remaining modular powers equals one. Every possible example therefore violates a necessary condition.

The loop parameters use 64-bit unsigned integers and never exceed $10^{10} + 1$. The only native squaring of an unreduced m occurs behind the guard $m \leq 10^4$ when testing $v = p$. The large polynomial and modular power are formed in GMP; near the upper boundary the modulus needs 67 bits and is not silently truncated to 64. The full execution receipt records the exact code hash, commands, types and counts. A separate 109-value check, including both sides of 2^{32} and the upper endpoint, compares this GMP arithmetic with Python's unbounded integers. Direct unsieved Python testing through 10^6 gives the same empty necessary candidate list. These latter checks are optional validation; the full exact bounded run and the retained Feit theorem are the proof premises. □

This recovers the authors' finite numerical remark, not a new unbounded answer about non-Desarguesian flag-transitive planes. The original Magma and GAP source code was not supplied and its complete cost is unknown. The complete current C++ file and its operational runner will be disclosed separately under the symmetric executable-code policy.

7 Literal source corrections and scope

The source's classification strategy and its retained inputs remain attributed to their authors. The present common interfaces replace repeated local arguments and make the acting-group conditions explicit. For clarity, the complete source map records the following corrections, in addition to the retained quoted results and recovered proofs.

1. Lemma 3.5 has a new complete proof at its full affine scope; its printed automatic-nesting and full-group-equality sentences are not used. This is a proof repair, not a theorem counterexample.

2. The induced groups in Propositions 3.7–3.8 contain $\mathrm{PSL}_2(q)$. Containment of $\mathrm{PGL}_2(q)$ is not necessary. The explicit $\mathrm{ASL}_2(3)$ example distinguishes them. The Proposition 3.9 conclusion survives: the order-four lift proof replaces its Sylow fixed-direction shortcut.

3. The WBS full group need not fail local point-primitivity: $W(8)$ has line action A_4. The uniform no-RWPRI conclusion remains true, with the pencil failure included. Its retained identification with the smallest Ree unital is unchanged.

4. The Hermitian blanket local-two-transitivity assertion is false when $\gcd(3, q + 1) = 3$ for the special-unitary group; at $q = 2$ even local primitivity fails for that group. The exact scalar-coset criterion replaces both affected converses.

5. The A_7 fifteen-point circle must be deleted from the source's primitive and three-transitive lists. The source's binary $G_2(2)'$ circle exclusion incorrectly treats a two-point line as thick; its primitive degree-63 pencil survives. One-dimensional primitive circles omitted from the standalone Proposition 4.3 are included explicitly. In Theorem 6.2 the explicit three-transitivity

antecedent makes its A_7 row empty; that redundant necessary row alone is not a counterexample to the theorem's implication. The binary exceptional case is already in the source's final affine G_2 branch, so its restored circle description does not enlarge that final union. Small alternating-group footnotes require the exact three- and four-point treatment given above.

6. Proposition 5.8 and the footnote after Proposition 5.6 need the three-point nonspread circle. The full acting-group converse in Theorem 6.1(1) needs line-restriction irreducibility, as witnessed by $2^4 : C_5$; the geometric uniqueness question is settled only at the already accepted one-dimensional locally primitive spread scope.

The source's stated $q = 2$ unitary/affine overlap is preserved: $\mathrm{ASL}_2(3)$ and $\mathrm{AGL}_2(3)$ occur on the nine-point affine plane/unital in both descriptions. This is an overlap of specified actions. We do not promote the source's tentative remark about intersections of families to a new exhaustive disjointness claim.

There are two required finite premises: the complete small $\mathrm{PSU}_3(3)$ maximal-subgroup calculation and the exact bounded Feit run. The WBS8 computation is optional validation of the uniform written proof. Operational compilation/render/checking files are inventoried separately. Source software is mentioned but not supplied; its full executable cost is unknown. Any later measurement must apply symmetric executable boundaries and retain that unknown, not count only our software as a compression factor. No Lean theorem is added and no end-to-end formalization is claimed. This private edition has no source-admission or numerical status before distinct review.

References

[1] F. Buekenhout, P.-O. Dehaye and D. Leemans, *RWPri and* $(2T)_1$ *flag-transitive linear spaces*, Beiträge zur Algebra und Geometrie **44** (2003), 25–46. All external classification locators above refer to this fixed edition's numbered statements and its complete original bibliography.

[2] *Local actions, invariant pencils and scalar quotients*, companion common paper in this private edition. It includes the full previously reviewed phase-four semilinear descent proof, with that prior work and its dependencies explicitly attributed.